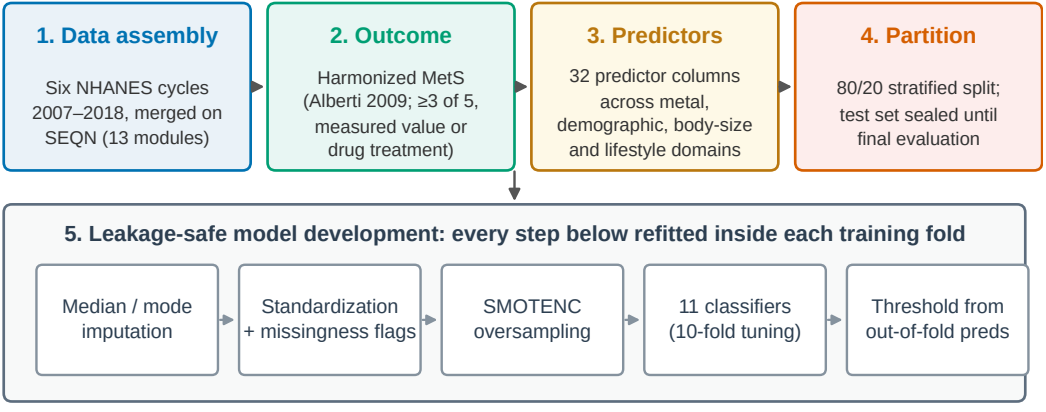
1. Data assembly
Six NHANES cycles 2007–2018, merged on SEQN (13 modules)
2. Outcome
Harmonized MetS (Alberti 2009; ≥3 of 5, measured value or drug treatment)
3. Predictors
32 predictor columns across metal, demographic, body-size and lifestyle domains
4. Partition
80/20 stratified split; test set sealed until final evaluation
5. Leakage-safe model development: every step below refitted inside each training fold
Median / mode imputation
Standardization + missingness flags
SMOTENC oversampling
11 classifiers (10-fold tuning)
Threshold from out-of-fold preds